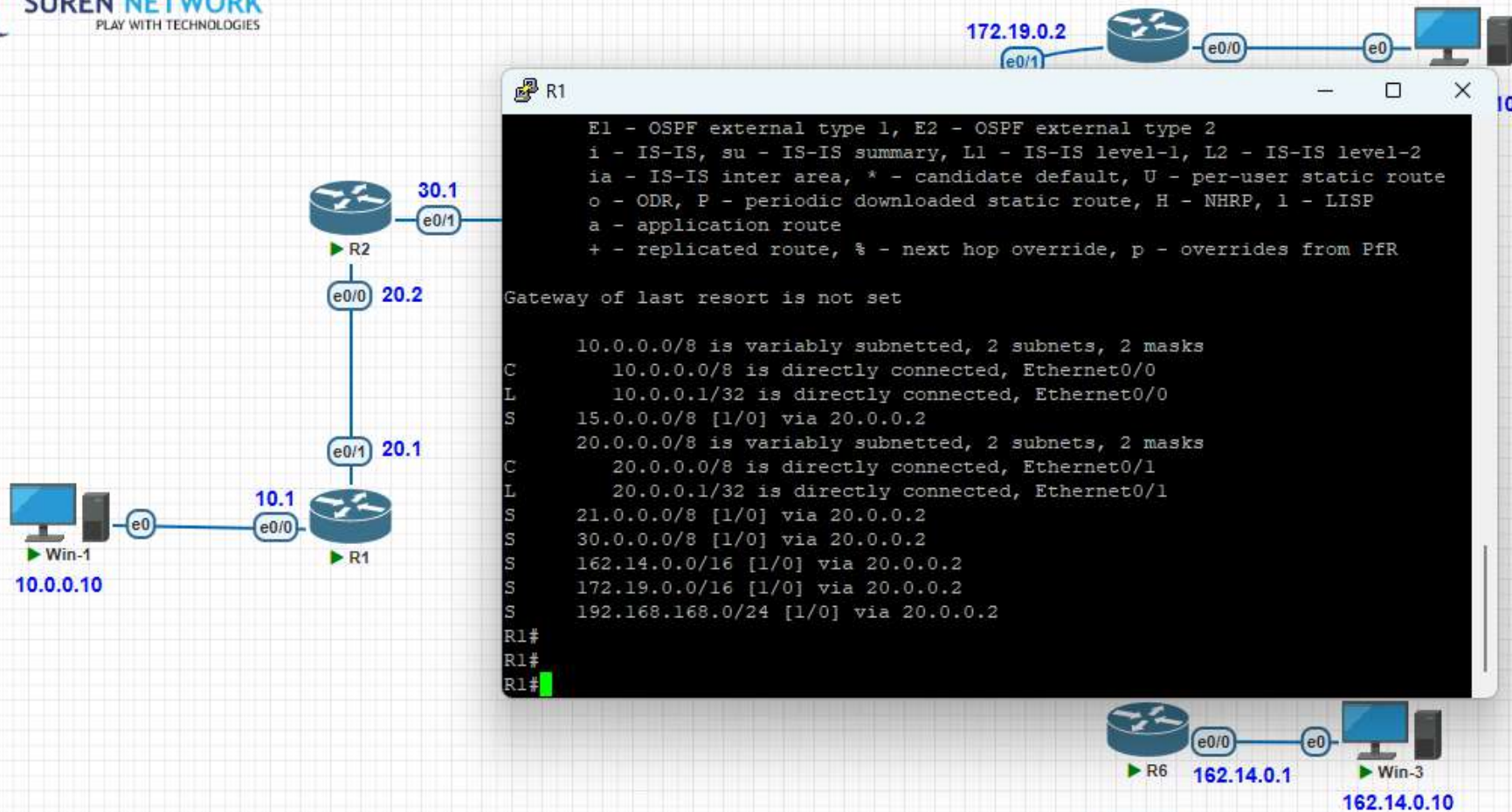




```
R1
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

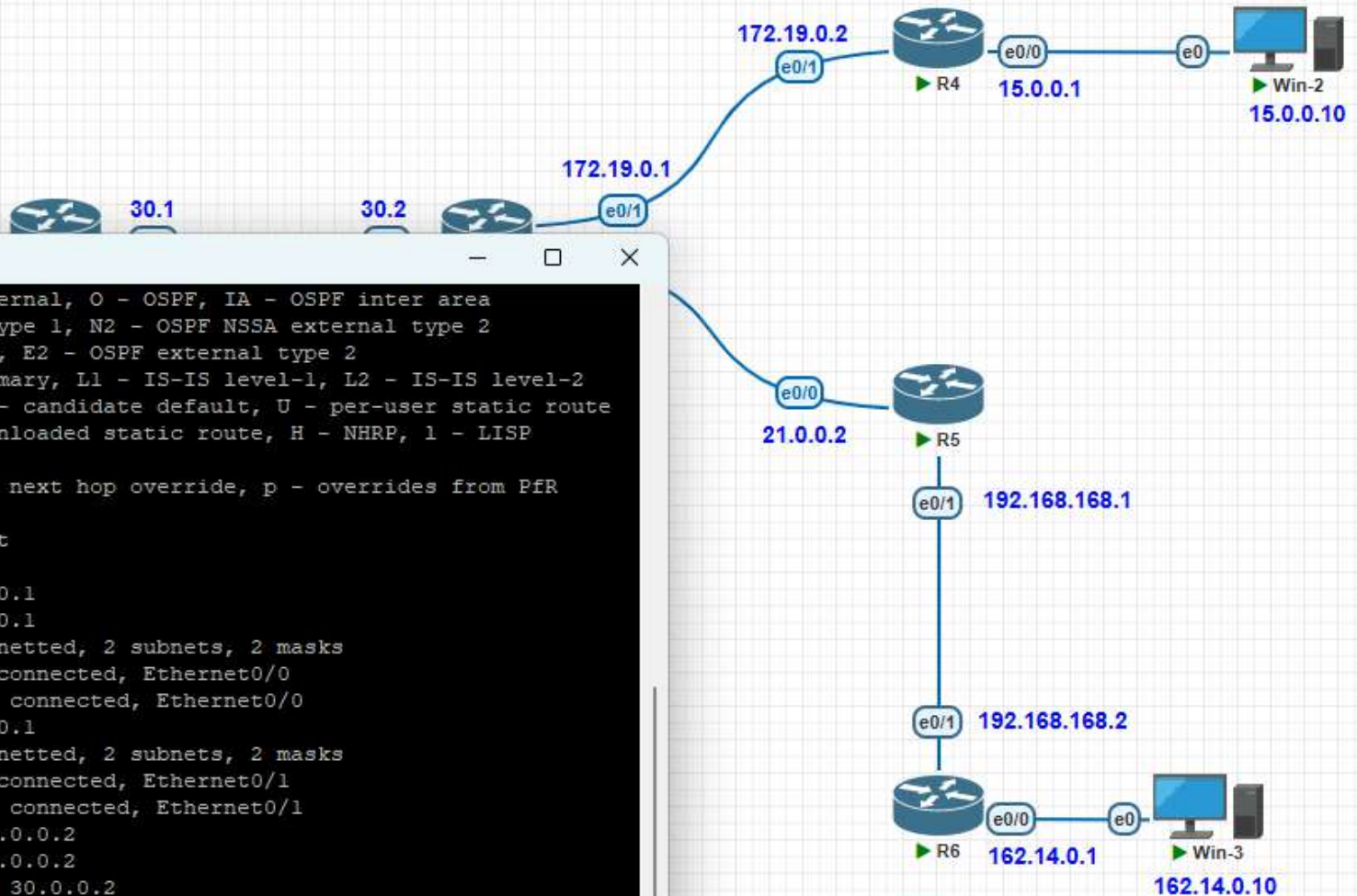
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
S    15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    21.0.0.0/8 [1/0] via 20.0.0.2
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    162.14.0.0/16 [1/0] via 20.0.0.2
S    172.19.0.0/16 [1/0] via 20.0.0.2
S    192.168.168.0/24 [1/0] via 20.0.0.2
R1#
R1#
R1#
```

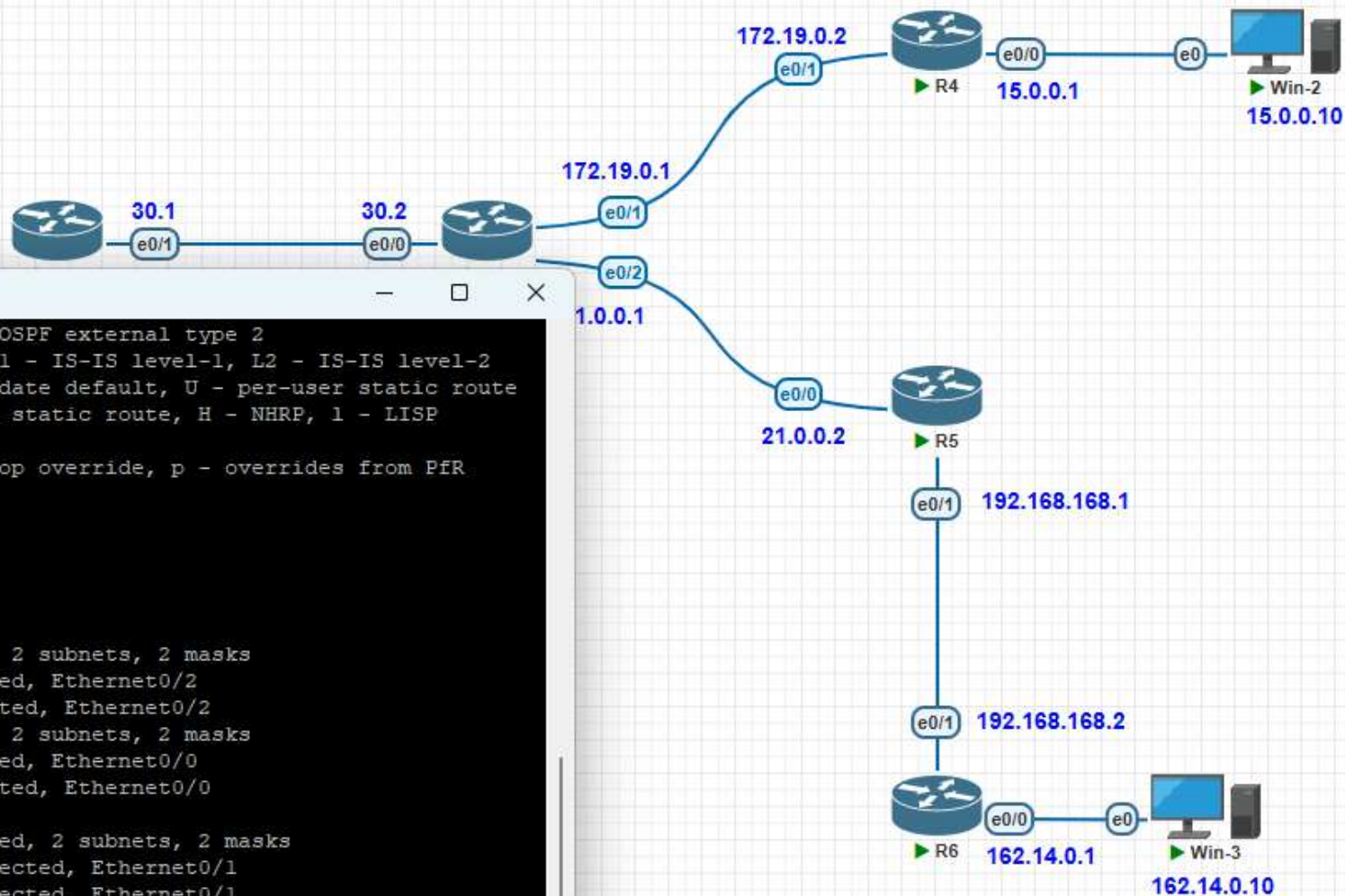
```

R2
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S   10.0.0.0/8 [1/0] via 20.0.0.1
S   15.0.0.0/8 [1/0] via 20.0.0.1
C   20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L   20.0.0.2/32 is directly connected, Ethernet0/0
S   21.0.0.0/8 [1/0] via 20.0.0.1
C   30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L   30.0.0.1/32 is directly connected, Ethernet0/1
S   162.14.0.0/16 [1/0] via 30.0.0.2
S   172.19.0.0/16 [1/0] via 30.0.0.2
S   192.168.168.0/24 [1/0] via 30.0.0.2
R2#
  
```





R3

E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

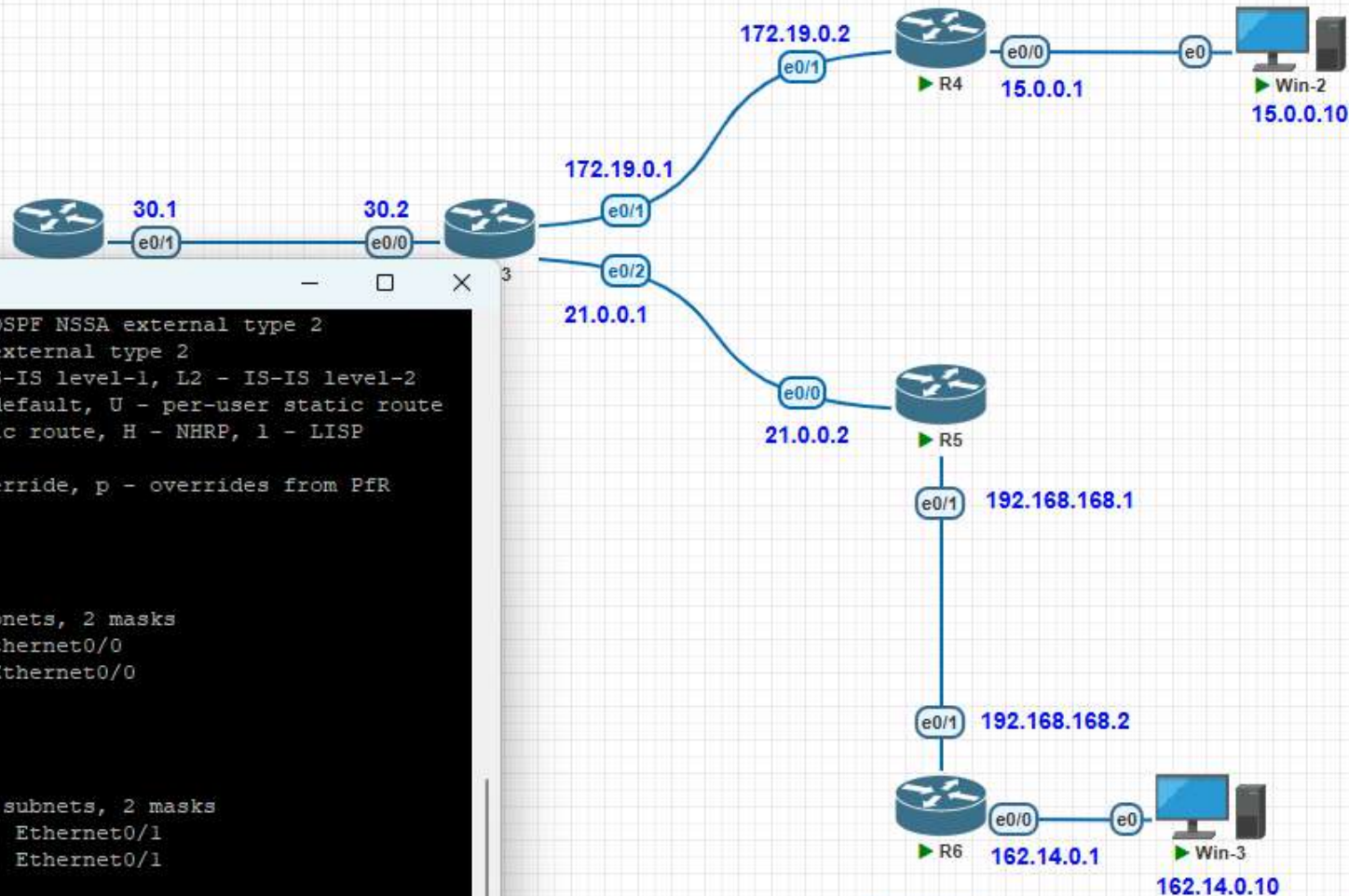
```

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
C    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
C    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
L    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
  
```

R3#

Physical

01 : 00



R4

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 172.19.0.1
S 15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/8 is directly connected, Ethernet0/0
L 15.0.0.1/32 is directly connected, Ethernet0/0
S 20.0.0.0/8 [1/0] via 172.19.0.1
S 21.0.0.0/8 [1/0] via 172.19.0.1
S 30.0.0.0/8 [1/0] via 172.19.0.1
S 162.14.0.0/16 [1/0] via 172.19.0.1
S 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.2/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 172.19.0.1
```

R4#

R4#

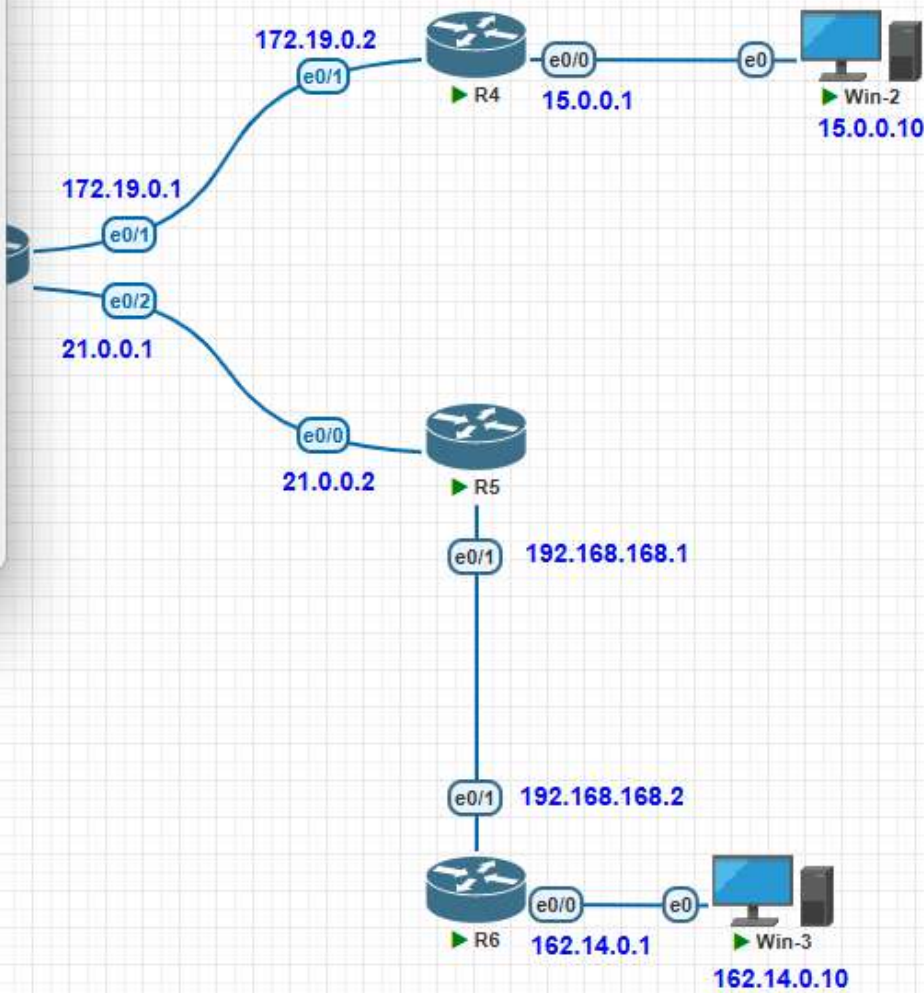
Physical

01 : 00


```
R5
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

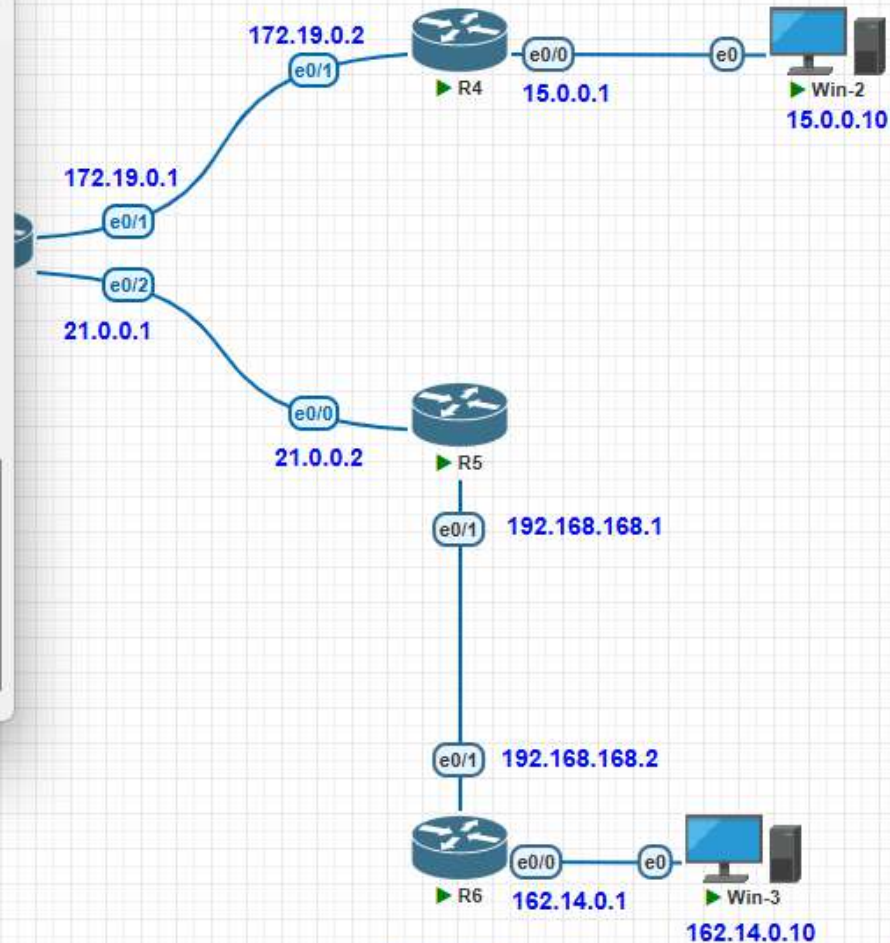
S   15.0.0.0/8 [1/0] via 21.0.0.1
S   20.0.0.0/8 [1/0] via 21.0.0.1
S   21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S   30.0.0.0/8 [1/0] via 21.0.0.1
S   162.14.0.0/16 [1/0] via 21.0.0.1
S   172.19.0.0/16 [1/0] via 21.0.0.1
S   192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#
```



```
R6
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

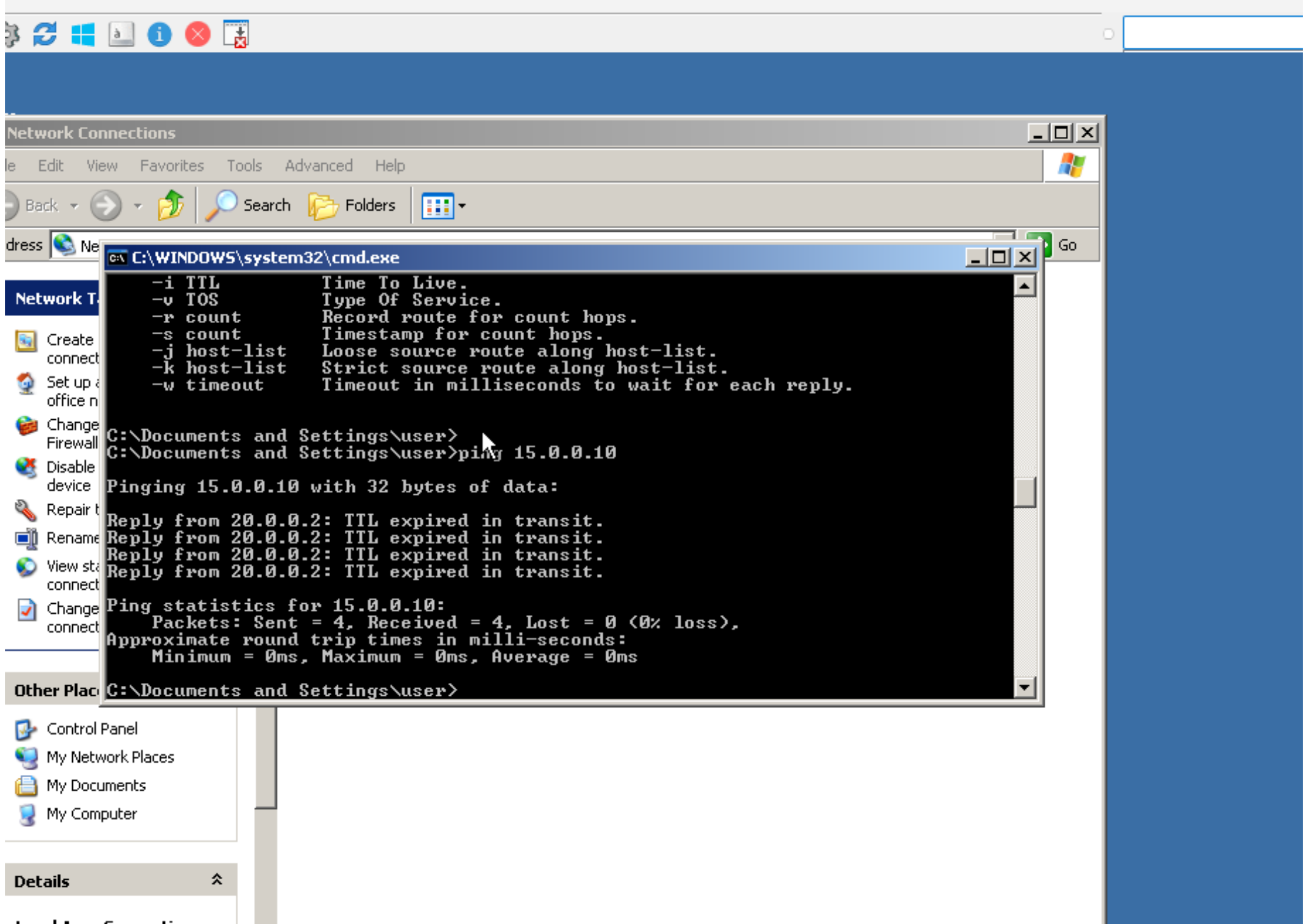
Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1
R6#
```



Physical

01 : 00 : 00



Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Stop Search Folders

Address Bar Go

Network Tasks

- Create a new network connection
- Set up a wireless network
- Change Firewall settings
- Disable a network device
- Repair a network connection
- Rename a network connection
- View status of a network connection
- Change network connection properties

Other Places to Look

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

```
C:\WINDOWS\system32\cmd.exe

-i TTL           Time To Live.
-v IOS           Type Of Service.
-r count         Record route for count hops.
-s count         Timestamp for count hops.
-j host-list     Loose source route along host-list.
-k host-list     Strict source route along host-list.
-w timeout       Timeout in milliseconds to wait for each reply.

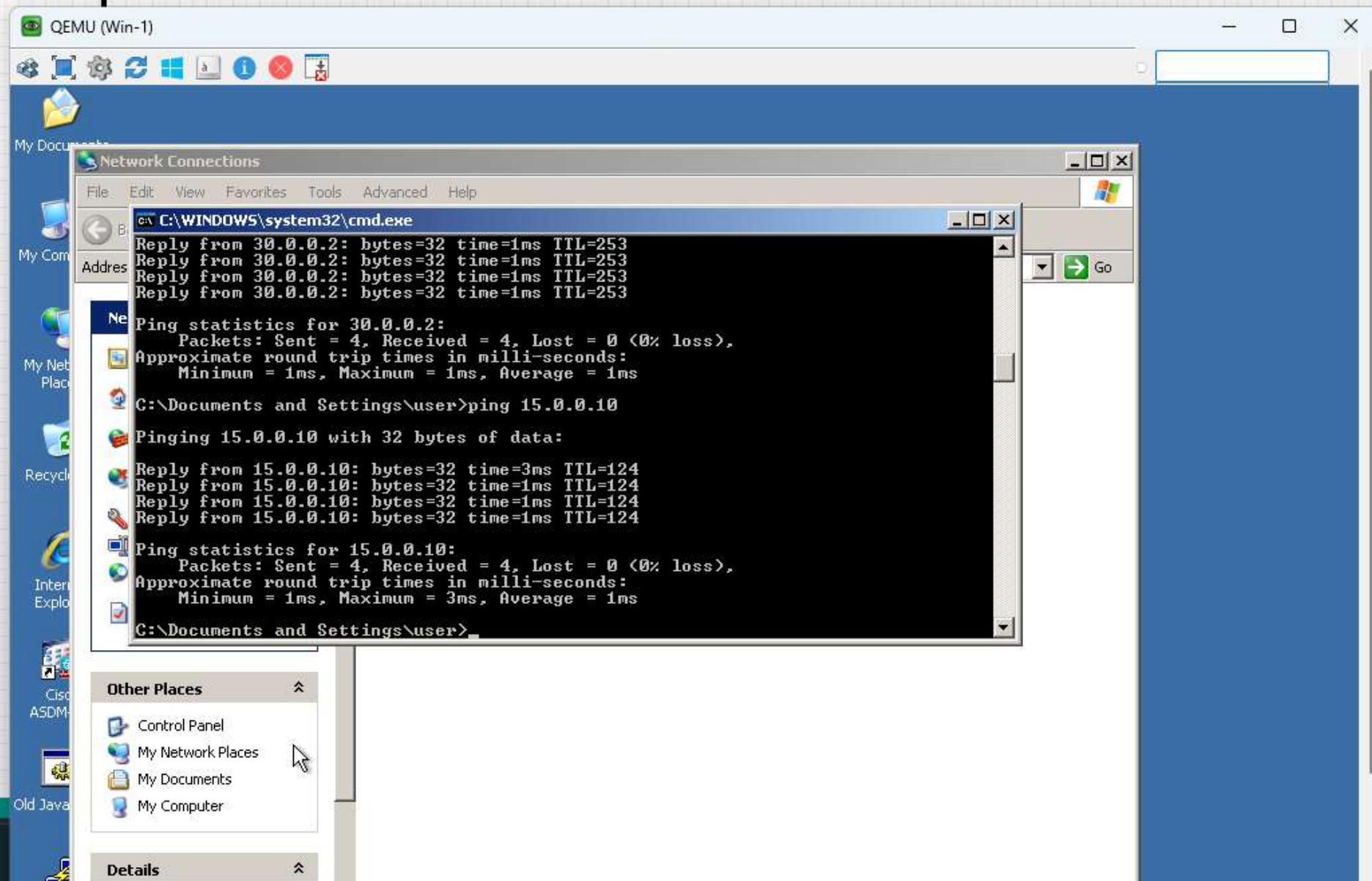
C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 20.0.0.2: TTL expired in transit.
Reply from 20.0.0.2: TTL expired in transit.
Reply from 20.0.0.2: TTL expired in transit.
Reply from 20.0.0.2: TTL expired in transit.

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

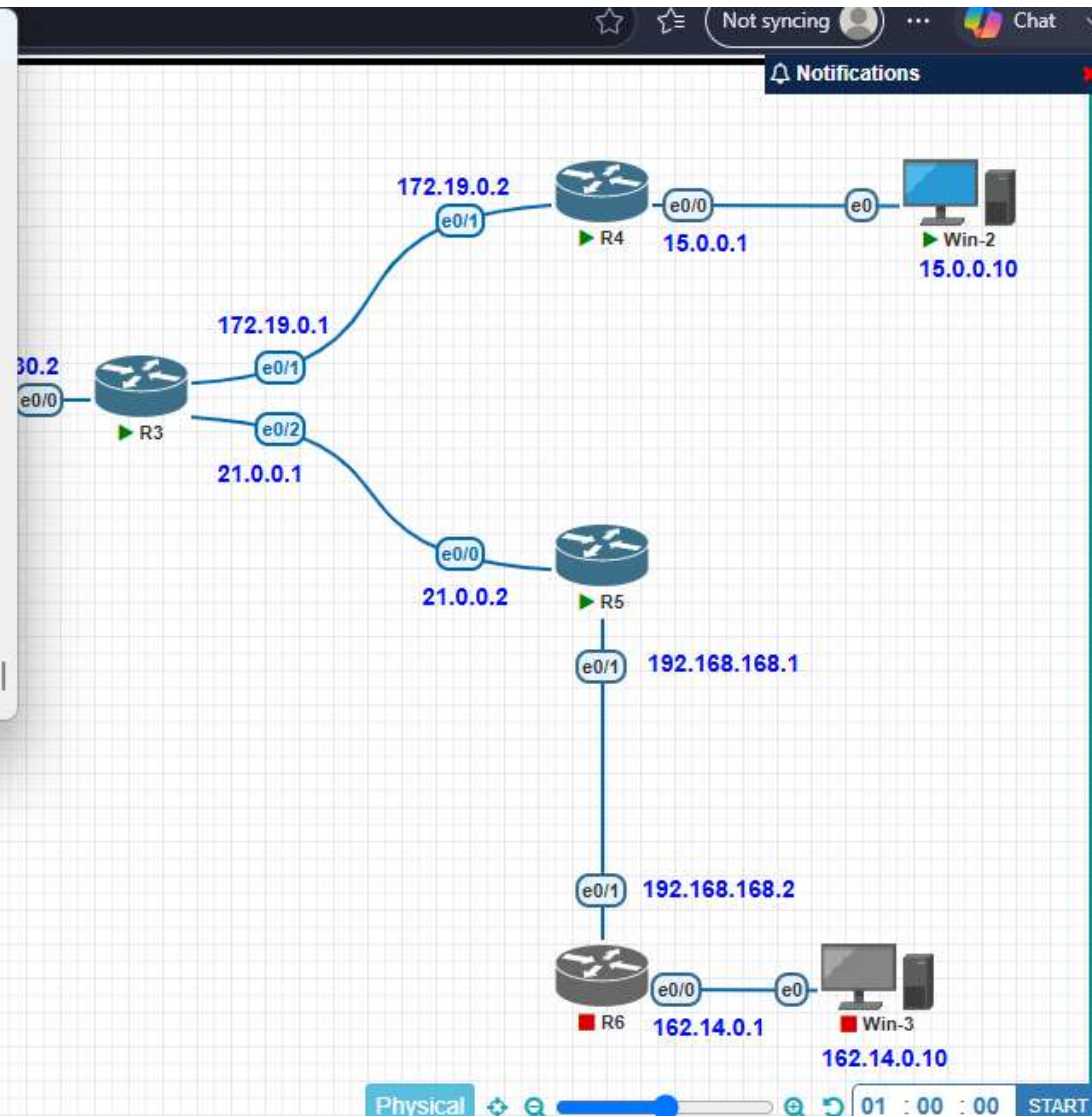
C:\Documents and Settings\user>
```


```
R5
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S   10.0.0.0/8 [1/0] via 21.0.0.1
S   15.0.0.0/8 [1/0] via 21.0.0.1
S   20.0.0.0/8 [1/0] via 21.0.0.1
S   21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S   30.0.0.0/8 [1/0] via 21.0.0.1
S   162.14.0.0/16 [1/0] via 21.0.0.1
S   172.19.0.0/16 [1/0] via 21.0.0.1
S   192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.1/32 is directly connected, Ethernet0/1
R5#
```



QEMU (Win-1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM

Old Java

PUT

Network Connections

File Edit View Favorites Tools Advanced Help

Address Bar

Go

Other Places

Details

Local Area Connection

C:\WINDOWS\system32\cmd.exe

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 15.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>ping 21.0.0.2

Pinging 21.0.0.2 with 32 bytes of data:

Reply from 21.0.0.2: bytes=32 time=2ms TTL=252
Reply from 21.0.0.2: bytes=32 time=2ms TTL=252
Reply from 21.0.0.2: bytes=32 time=2ms TTL=252
Reply from 21.0.0.2: bytes=32 time=2ms TTL=252

Ping statistics for 21.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>

Not syncing

Notifications

15.0.0.1

15.0.0.10

192.168.168.1

192.168.168.2

162.14.0.1

162.14.0.10

01 : 00 : 00



My Documents



My Computer



My Network Places



Recycle Bin



Internet Explorer



Cisco ASDM



Old Java



Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Stop Search Folders

Address **C:\WINDOWS\system32\cmd.exe** Go

```
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.1

Pinging 162.14.0.1 with 32 bytes of data:

Reply from 162.14.0.1: bytes=32 time=1ms TTL=255
Reply from 162.14.0.1: bytes=32 time<1ms TTL=255
Reply from 162.14.0.1: bytes=32 time=1ms TTL=255
Reply from 162.14.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 162.14.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>
```

Other Places

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

R6

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

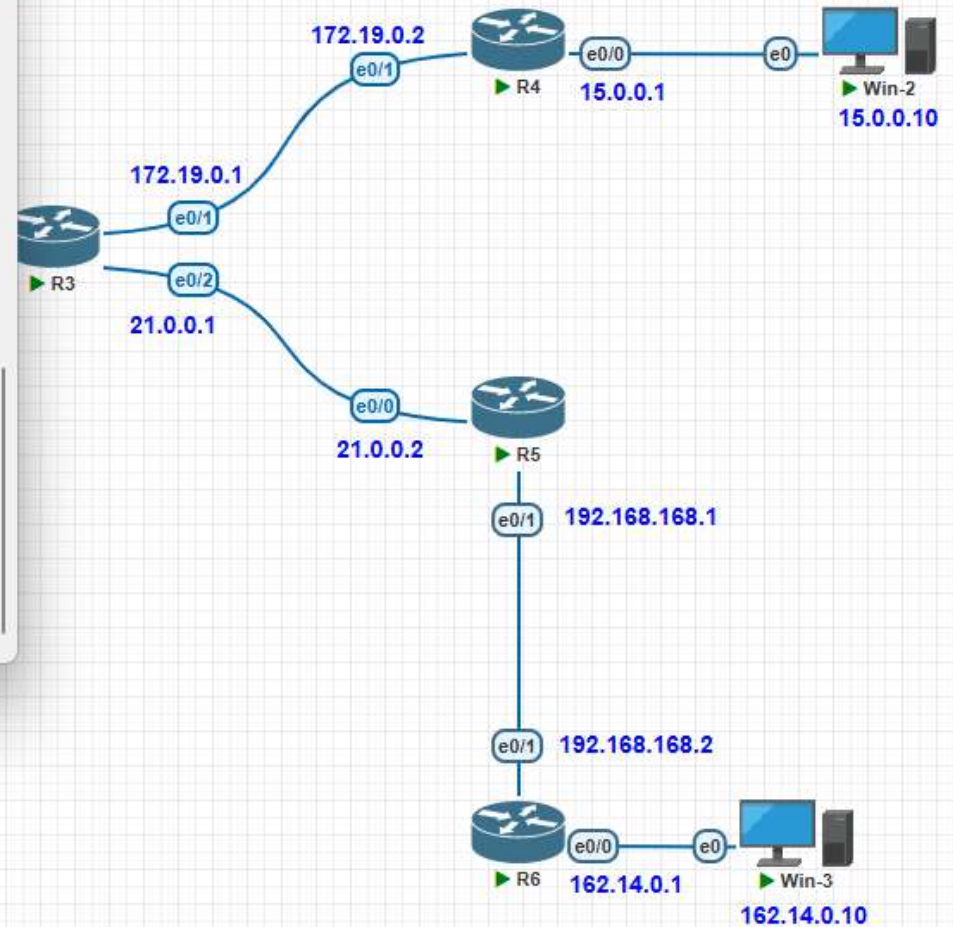
Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
C 162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
L 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
C 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
L 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1
```

R6#

R6#

10.0.0.10



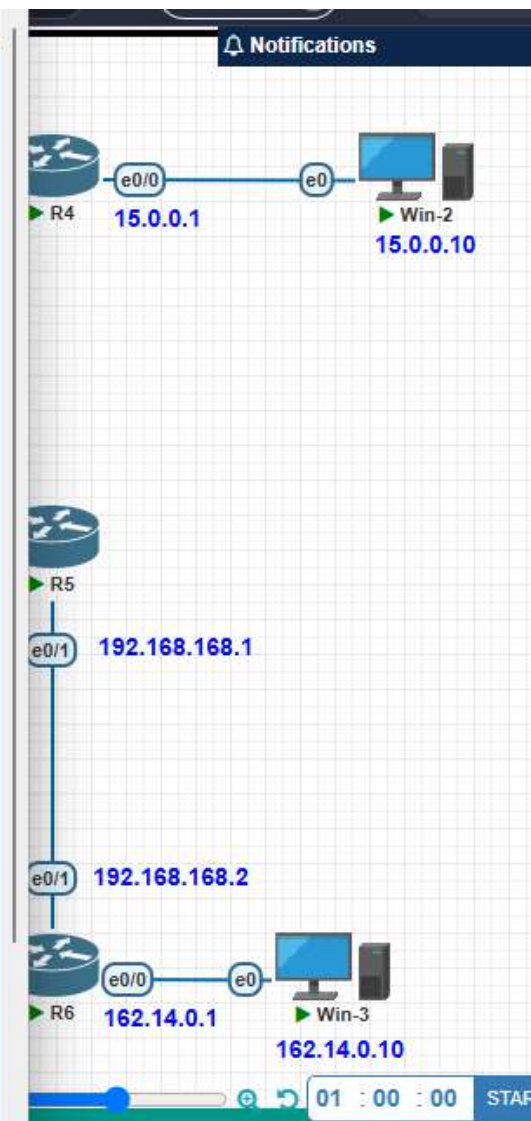
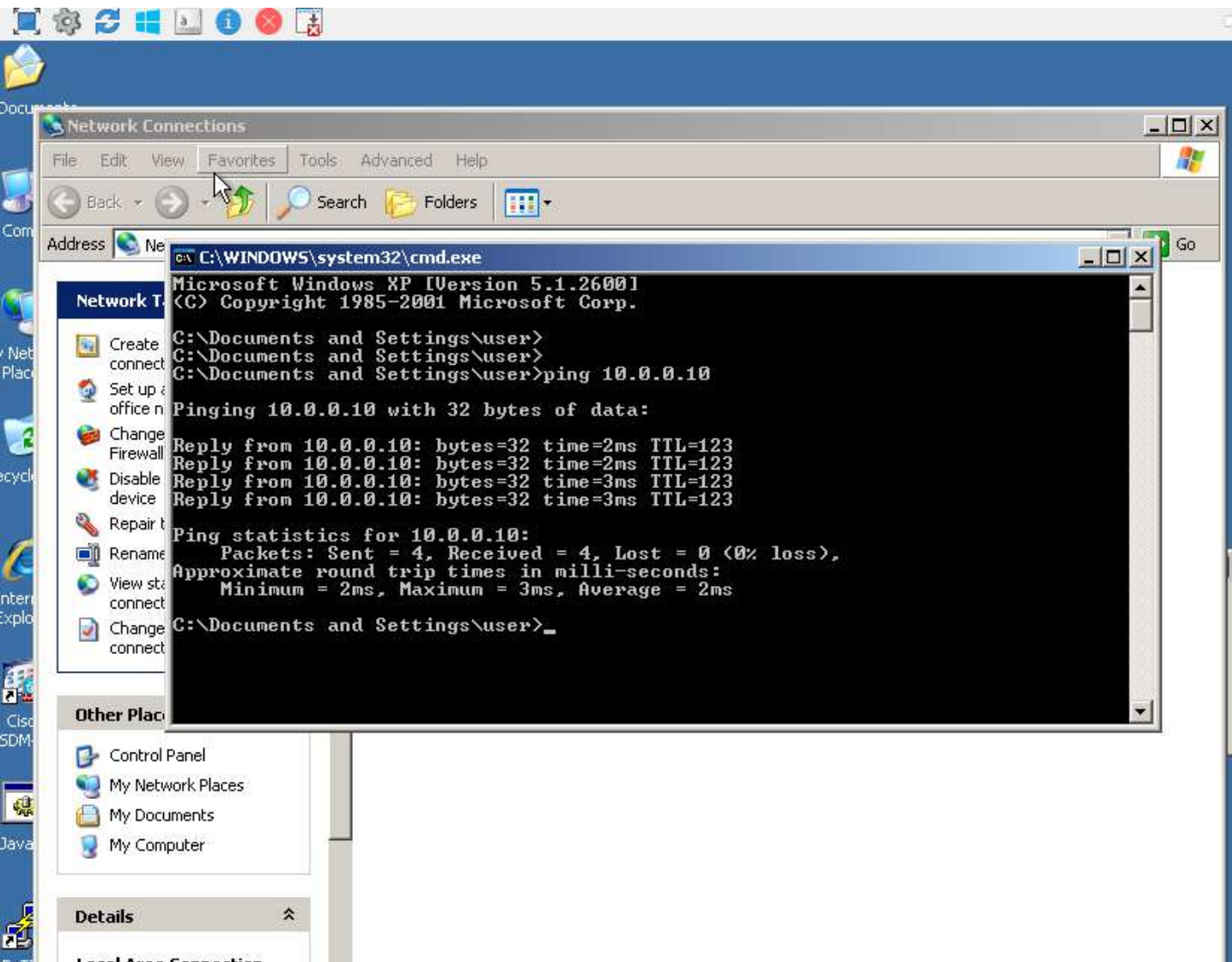
Physical

+

-

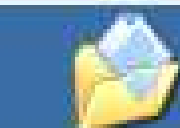
01 : 00 : 00

START



https://labs.surenetwork.com/legacy/topology

QEMU (Win-3)



My Documents



My Computer



My Network Places



Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Stop Search Folders

Address New

Network Tasks

- Create a new network connection
- Set up a wireless office network
- Change network settings

C:\WINDOWS\system32\cmd.exe

```
Ping statistics for 172.19.0.2:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)  
  
C:\Documents and Settings\user>ping 15.0.0.10  
  
Pinging 15.0.0.10 with 32 bytes of data:  
  
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
```


QEMU (Win-3)

Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Search Folders

Address [C:\WINDOWS\system32\cmd.exe](#) Go

Network T

- Create connect
- Set up office n
- Change Firewall
- Disable device
- Repair t
- Rename
- View sta connect
- Change connect

Other Place

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

```
C:\WINDOWS\system32\cmd.exe
Ping statistics for 172.19.0.2:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
```

